Optimizing Inventory Management and Sales Performance in a Retail Pharmacy using Data-Driven Techniques

A Midterm report for the BDM capstone Project

Submitted by

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1 Executive Summary:

Ganesh Medicals is a small B2C retail pharmacy located at Bajla Chowk, Deoghar.The pharmacy caters to both regular walk-in and prescription based customers, maintaining a broad range of medicines. Despite maintaining a digital sales and purchase records, the business faces three interconnected operational challenges: inaccurate inventory forecasting leading to stock imbalances, limited alignment of stock with seasonal demand shifts and supplier related procurement constraints.

These issues arise from internal factors such as heuristic-based purchasing decisions and limited analytical tracking of fast and slow-moving items, as well as external factors including fluctuating local demand patterns and variable supplier lead times. Since explicit expiry dates are not available in the dataset, expiry-linked losses are assessed using proxy measures such as sales velocity and batch-level retention duration. These issues are analyzed using descriptive statistics, item-level aggregation, value-based ABC analysis based on sales contribution, and proxy-based inventory ageing indicators.

Seasonality-driven demand misalignment is influenced internally by fixed procurement cycles and the absence of demand trend monitoring, and externally by short-term seasonal and climate-related demand variations. Month-wise sales aggregation and trend analysis are applied to evaluate intra-year demand fluctuations and assess the alignment between sales patterns and procurement timing.

Supplier variability and procurement constraints originate internally from dependence on a limited set of suppliers and bulk ordering pratices, and externally from supplier availability fluctuations and trade discount structures. Supplier-wise purchase aggregation and variability analysis are used to examine procurement concentration and irregularity.

Microsoft Excel is used for data cleaning, normalization, aggregation, metadata documentation, and ABC classification, while Python libraries such as Pandas and Matplotlib support sales velocity computation, batch retention analysis, trend visualization, and comparative analysis. The study is structured over an eight to ten-week period, covering data preparation, descriptive analysis, problem-wise analytical evaluation, and synthesis of interim findings.

2 Proof of Originality of Data:

Primary data was collected from the pharmacy, which was recorded from period April, 2025 to 5-Oct, 2025 for sales and from April, 2025 to 12-Nov-2025 for inventory.

The pharmacy owner provided authorized access to the billing software database. The data was exported as excel files directly from the POS system and inventory database, ensuring automated data capture accuracy.

Data authenticity is evidenced through official letter from pharmacy owner, photos of the shop and short interaction video with the owner discussing business challenges and granting data access has been attached to the drive link below:

LINK:
(Please copy paste the link if it’s not clickable)

3 Metadata:

3.1 Metadata: sales dataset

	Date	Bill No#	Type	Party Name/Mode of payment	Item Name	Batch	Qty	Rate	Amount	MRP	MRP Amt	Compnay Name	Sale Type
0	1-Apr-25	R000001	Sale	CASH	DNS 0.45% 1*500	4HB21005	1.0	181.50	181.50	181.50	181.50	ZZZZZZ 2943	L
1	1-Apr-25	R000001	Sale	CASH	POTASSIUM CHLORID INJ 1*10ML 1*10ML	KP1307535	1.0	26.70	26.70	26.70	26.70	HINDUSTAN PHARMACEUTICALS	L
2	1-Apr-25	R000001	Sale	CASH	TAXIMAX-1500MG INJ* 1*1VIAL	24460970	1.0	93.50	93.50	93.50	93.50	ALKEM LABORATORIES LTD.	L
3	1-Apr-25	R000001	Sale	CASH	NS 100ML 1*100ML	A4200	1.0	22.03	22.03	22.03	22.03	AISHWARYA	L
4	1-Apr-25	R000001	Sale	CASH	MIKACIN 250Mg INJ* 1*1VIAL	BLE240225	1.0	61.30	61.30	61.30	61.30	ARISTO PHARMACEUTICALS PVT.LTD	L

The sales dataset captures transaction level information for medicines sold by the pharmacy for the study period 1, April, 2025 to 5, October, 2025. It captures key transactional details required to study inventory performance, seasonal demand shifts, and customer buying behavior.

Data Composition:

The sales dataset comprises of 65530 entries and 13 columns. Each record represents an individual sales transaction and includes attributes related to transaction like date, unique bill no. for every transaction, product identification, quantity pricing value and product manufacturer details .This dataset structure supports optimizing inventory via product-level sales velocity analysis, revenue tracking, and temporal demand pattern identification .

Key columns include:

- **Date:** Transaction date enabling time-series analysis and seasonal decomposition.
- **Item Name:** Pharmaceutical product identifier with quantity, dosage strength and form. (e.g., ―SUMO L 250MG SYP* 1*60ML‖)
- **Batch:** Manufacturing batch number linking purchases to sales for turnover analysis.
- **Qty:** Units sold per transaction, with negative values indicating returns or stock outs.
- **Party Name/Mode of Payment:** Payment method (CASH, CREDIT) or customer type, with ―shortfall‖ notation marking unfulfilled demand or marked with ―S/Re‖ for customer returns. This column serves a dual purpose, capturing both payment methods for retail transactions and customer identifiers for doctor credit transactions (surgical/specialty medicines sold on credit), enabling segmentation analysis.
- **Amount:** Actual transaction value for revenue analysis.
- **MRP:** Maximum Retail Price for margin calculation and pricing compliance verification.

Data Integrity & Anomalies:

The Anomalies in this dataset include:

- **Negative Quantities:** in Qty column depicts returns and stock outs based on "shortfall" notation in Party Name field. Due to presence of wide ranges (-56 to 150) may require sanity check and outlier identification.
- **Daily Aggregate rows**: Dataset contained daily summary rows with totals, on removal of these had led to empty rows and null values that must be removed during cleaning to prevent errors.

3.2 Metadata: procurement dataset

	Date	Bill No#	Type	Party Name	Item Name	Batch	Qty	Free Qty	Rate	Discount	GST %	Tax Amount	Amount	MRP	MRP Amt	Compnay Name	Sale Type
0	2025-04-01	15	Purc	B.K.ENTERPRISES DEOGHAR	ONDEM INJ 2ML* 1*2ML	OND24333SR	25.0	25.0	9.54	11.93	12.0	27.18	253.75	13.35	333.75	ALKEM LABORATORIES LTD.	L
1	2025-04-01	15	Purc	B.K.ENTERPRISES DEOGHAR	SUMO L 125MG SYP* 1*60ML	24491039	18.0	2.0	29.06	26.15	12.0	59.64	556.57	40.68	732.24	ALKEM LABORATORIES LTD.	L
2	2025-04-01	15	Purc	B.K.ENTERPRISES DEOGHAR	FLAGYL 400MG TAB* 1*15TAB	FLA24075	600.0	0.0	18.24	36.48	12.0	83.18	776.30	25.53	1021.20	PIRAMAL	L
3	2025-04-01	2	Purc	CASH	AAAA	NaN	2.0	0.0	14500.00	0.00	0.0	0.00	29000.00	15000.00	30000.00	APPLE I	L
4	2025-04-01	2	Purc	CASH	AAAA	NaN	2.0	0.0	14500.00	0.00	0.0	0.00	29000.00	15000.00	30000.00	APPLE I	L

The purchase dataset captures transaction-level stock information for medicines acquired by the pharmacy from the suppliers and distributors during the study period from April 1, 2025 to November 12, 2025. Each record corresponds to a purchase transaction and includes attributes related to date, supplier identification, product details, batch numbers, quantities regular and free(promotional), unit pricing, trade discounts, tax components (GST percentage and amount), and maximum retail price information. This dataset extends 38 days beyond the sales period to capture forward procurement planning and seasonal anticipation strategies.

Data Composition:

This dataset encapsulates key procurement details including Transaction Date, Supplier/Party Name, Item Name, Batch Number, Purchase Quantity, Free Quantity (trade schemes), Unit Rate, Discount Percentage, GST components, and MRP, crucial for understanding supplier performance, cost structures, and inventory inflow patterns.

Key columns include:

- **Date**: Purchase transaction date enabling temporal procurement pattern analysis
- **Party Name**: Supplier/distributor identifier for supplier performance evaluation
- **Item Name**: Product description requiring standardization for matching with sales data
- **Batch**: Manufacturing lot number for batch-level tracking and FIFO/FEFO analysis
- **Qty**: Units purchased per transaction
- **Free Qty**: Complimentary or promotional units received as trade benefits
- **Rate**: Purchase price per unit before discount
- **Discount**: Trade discount percentage offered by supplier
- **Amount**: Net purchase cost after discount, before tax
- **GST %** and **Tax Amount**: Tax components for compliance and total cost calculation
- **MRP**: Maximum Retail Price indicating potential profit margins.

Data Integrity & Anomalies:

- **Negative Quantities:** in Qty column depicts returns based on ―S/Re‖ notation in Type field. Due to presence of wide ranges (-90 to 1000) may require sanity check and outlier identification
- **Daily Aggregate rows**: Dataset contained daily summary rows with totals, on removal of these had led to empty rows and null values that must be removed during cleaning to prevent errors.

- **Item Name Formatting Variations**: Inconsistent abbreviations ,dosage or quantities results in the same medicine appearing under multiple item names (e.g. ―TAB‖ vs ―TABLET‖, ―INJ‖ vs ―INJECTION‖, ―SUMO L 125MG SYP* 1*60ML‖ vs ―SUMO L 250MG SYP* 1*60ML‖) require text standardization for reliable matching with purchase dataset.
- **Missing Batch Numbers:** Some transactions lack batch information, potentially indicating data entry gaps or products from legacy inventory systems. These entries cannot participate in batch-level turnover calculations.

4 Descriptive Statistics: Sales dataset

4. 1: Temporal Distribution Summary

Metric	Value
Study Duration	188 days (Apr 1- Oct 5, 2025)
Total Transactions	65343 (after cleaning)
Average Daily Transactions	347.56
Peak Transaction Day	09-09-2025(by quantity)
Weekday Transactions	49160 (75.2%)
Weekend Transactions	16183(24.76%)

Business Insight: These descriptive statistics show that pharmacy handles a consistently high transaction volume, averaging around 348 transactions /day over the 188-day period, indicating steady demand that requires reliable inventory planning. Weekday sales dominate, with over 75% of transactions occurring from Monday to Friday, suggesting that procurement and staffing decisions should prioritize weekday peaks rather than weekends. The presence of a clear peak day (09-09-2025) also hints at potential seasonality or event-driven surges.

4.2: Sales Volume Statistics

Statistic	Value	Formula
Total Units Sold (Gross)	105787.4	Sum of positive Qty
Total Units Returned	2913.8	Absolute value of return Qty
Total Stockout Quantity	467.5	Absolute value of stockout Qty
Net Units Sold	102873.56	Gross - Returns
True Demand	106254.9	Gross Sales + Stockouts
Mean Qty per Transaction	1.65	Average of positive Qty

Median Qty	1	Median of positive Qty
Standard Deviation	1.7	StDev of positive Qty
Coefficient of Variation	105%	(Std Dev / Mean) × 100
Return Rate	2.75%	(Returns / Gross Sales) × 100
Stockout Rate	0.43%	(Stockouts / True Demand) × 100

Business Insight: The pharmacy fulfilled a substantial demand of over 1.06 lakh units during the study window, but about 2.75% of units were returned and 0.43% could not be served due to stockouts, directly tying into inventory inefficiency and expiry-linked losses .A high coefficient of variation (105%) indicates highly variable basket sizes, which makes rule-of-thumb purchasing, indicating that rule-of-thumb purchasing practices are insufficient in such a volatile demand environment. The presence of measurable return and stockout rates further suggests scope to refine reorder points and batch sizing so that true demand is captured more consistently and avoidable sales reversals or losses are minimized.

4.3: Revenue and Product Concentration

Metric	Value	Relevance
Total Revenue (Gross)	Rs.9920890.660000002	Overall business performance
Revenue Lost to Returns	Rs.177819.06	Quality/expiry indicator
Revenue Lost to Stockouts	Rs.63991.71	Opportunity cost of understocking
Net Revenue	Rs.9743071.600000001	Actual income after returns
Average Transaction Value	Rs.787.2473147119505	Typical customer spend
Median Transaction Value	Rs. 582.565	Typical purchase size
Total MRP Value	Rs. 9920890.660000002	Maximum potential revenue
Revenue Realization Rate	100.0	(Actual/MRP) × 100
Unique Products (SKUs)	766	Profitability indicator
Top 20% Products (A-items)	169 product items contributing ~80% revenue	Pareto principle applies

Business Insight: The pharmacy generated nearly ₹99 lakh in gross revenue over the study period, with returns and stockouts causing only modest revenue leakage. Revenue is highly concentrated: 169 SKUs (about 22%) contribute roughly 80% of sales, highlighting strong scope for focused inventory and availability management on these key products.

Descriptive Statistics: Procurement dataset

4.4: Procurement Metrics

Metrics	Value	Business Relevance
Total Units Purchased	118311.5	Inventory inflow
Total Free Units	23796.0 (20.11%)	Effective cost reduction through supplier-provided quantity incentives
Total Procurement Cost	Rs.7272218.28 (including gst and discount)	Capital investment in inventory
Total Procurement returned Units	244.0	Indicator of procurement inefficiency and demand–supply mismatch
Total Procurement returned Value	20567.690000000006	Direct financial impact of excess or misaligned procurement
Average Discount %	4.038	Typical supplier terms
Discount Std Dev %	1.808	Variability indicator
Top 3 Suppliers	KEDIA DRUG AGENCY DEOGHAR, R.P DISTRIBUTORS DEOGHAR, B.K.ENTERPRISES DEOGHAR contributing **48.22 %** of total procurement value.	Supplier concentration

Business Insight: Procurement heavily relies on three key suppliers, who together accounts for nearly half of total purchase value, creating both negotiation opportunities and dependency risk. A high share of free units (20.11%) and average discounts of 4.038% with notable variability suggest that trade schemes strongly influence buying decisions, potentially distorting alignment with true demand.

Descriptive Statistics: Cross-Dataset Comparison

Table4.5: Sales vs Purchase Alignment

Dimension	Sales Data	Purchase Data	Misalignment Indicator
Date Range	1 Apr-5 Oct(188 days)	1 Apr-12 Nov(226 days)	Purchase extends 38 days
Monthly Avg Units	17631	16866	Ratio: 1.04
Monthly Avg Value	1653481.776666667	1035950.0842857143	
Peak Month (Sales)	9 (23194.6)	-	Tests seasonal procurement response
Peak Month (Procurement)	-	9(22497.8)	

Business Insight: Sales and purchase volumes are broadly aligned in units, but procurement continues 38 days beyond the analyzed sales window, indicating a risk of overstock relative to observed demand. The shared peak in September confirms seasonality, yet longer procurement coverage without matching sales visibility may aggravate expiry and working-capital pressures (Problems 1.1 and 1.2)

5 Detailed Explanation of Analysis Process/Methods

5.1 Methods for Problem Statement 1: Inventory Inefficiency and Expiry-Linked Losses

5.1.1 ABC Classification with Sales Velocity:

- **Definition:** Products are categorized based on revenue contribution (A: top 20% contributing -80% of revenue; B: middle 30% contributing -15%; C: bottom 50% contributing -5%) and combined with sales velocity measured as units sold per day.
- **Justification**: This method is appropriate for Problem Statement 1 as it simultaneously identifies slow-moving products prone to expiry risk and high value products where stockouts lead to significant revenue loss. Combining value based classification with sales velocity prevents reliance on a single metric and enables differentiated inventory control, where high-value items require tight

availability management while low value, slow moving items warrant stock reduction.

- **Expected output:** Combining value-based classification with sales velocity prevents reliance on a single metric and enables differentiated inventory control, where high-value items require tight availability management while low-value, slow moving items warrant stock reduction.

5.1.2 Stockout Impact Analysis:

- **Definition:** Compute unfulfilled demand per SKU using: (Stockout Qty/(Sales Qty+Stockout Qty))×100
- **Justification:** More informative than sales-only views, which hide unmet demand when shelves are empty. Leverages explicit —shortfall‖ entries to reconstruct true demand rather than just observed sales. More precise than qualitative or anecdotal stockout complaints.
- **Expected output:** A list of SKUs with high unfulfilled-demand percentages marked as chronic under-stock items, along with total revenue loss, to prioritize corrective action.

5.2 Methods for Problem Statement 2: Seasonality-Driven Demand Misalignment

5.2.1 Month-over-Month Growth Comparison (Sales vs Purchase)

- **Definition:** Calculate percentage change between consecutive months for both sales and purchases:
- MoM Growth = ((Month_N - Month_N-1) / Month_N-1) × 100
- **Justification:** This method directly measures how quickly procurement responds to changing demand by comparing month-over-month growth in sales and purchases with a clear, percentage-based metric that stakeholders can easily interpret. Because it uses growth rates rather than raw values, it remains comparable across high and low volume months, works even with only a single year of data , and offers more rigor than visual inspection alone when paired with correlation between sales and purchase growth.
- **Expected output:** Correlation coefficient between sales and purchase MoM growth (low correlation = poor alignment) and identification of months where

sales–purchase growth differs by more than a set threshold , as concrete evidence of seasonality misalignment.

5.2.2 True Demand Reconstruction

- **Definition:** True Demand = Sales Quantity + |Stockout Quantity|
- **Justification:** This method adjusts for the fact that stockouts cap observed sales, so recorded volumes understate actual demand in peak periods. By adding back documented shortfalls, it reveals the real seasonal intensity that procurement should plan for, using information that most typical datasets do not capture. This makes it more accurate than relying on sales-based seasonality alone or ignoring stockout data, where demand is implicitly treated as equal to what was available rather than what customers actually wanted.
- **Expected output:** True demand reconstruction shows that observed sales understate actual demand during **seasonal transition months**, rather than peak sales months. After accounting for stockouts, true demand exceeds recorded sales by approximately **0.9% in April** and **2.4% in July**, indicating procurement stress during periods of demand acceleration rather than during peak demand itself.

5.3 Methods for Problem Statement 3: Supplier Variability

5.3.1 Effective Rate Calculation

- **Definition:** True cost per unit after incorporating discounts AND free quantities:
- **Effective Rate = (Rate × (1 - Discount%) × Qty) / (Qty + Free Qty)**
- **Justification:** It captures the hidden value of free-quantity schemes that can significantly alter actual costs even when discount percentages look similar. By normalizing transactions with different combinations of prices, discounts, and free units, it allows consistent comparison of suppliers and highlights cost differences that unit price or discount percentage alone would miss. Unlike simple averaging, effective rate weights costs by actual quantities procured, so high-volume purchases have appropriate influence and supplier-level variability linked to Problem Statement 3 becomes visible.
- **Expected output:**The method is expected to uncover meaningful differences in true per-unit cost across suppliers once discounts and free units are accounted for,

providing quantitative evidence of supplier heterogeneity and potential procurement inefficiencies that can be targeted in later optimization.

5.3.2 Supplier Performance Scorecard

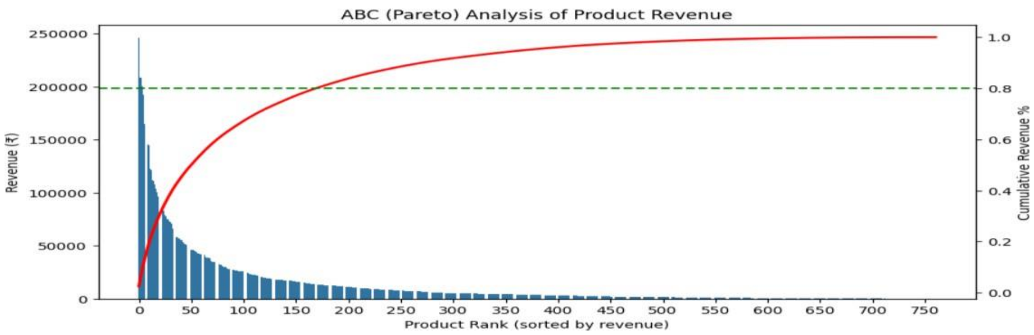
- **Definition:** Build a composite supplier score using multiple criteria with weights, for example: discount % (30%), free-quantity % (20%), price competitiveness (30%)
- **Justification:** It provides a holistic, data-driven evaluation instead of relying on a single metric like price or informal judgement. The weighted structure reflects business priorities (cost and reliability) and produces an objective, repeatable ranking that directly supports procurement decisions under Problem Statement 3.
- **Expected output:** A 0–100 score for each supplier, highlighting the top performers
- as strategic partners and flagging weaker suppliers for renegotiation or replacement.

6 Results and Findings

6.1 Findings for PS1: Inventory Inefficiency

Finding 6.1.1: Pareto Concentration with Stockout Overlay

Graph 6.1.1: ABC Classification

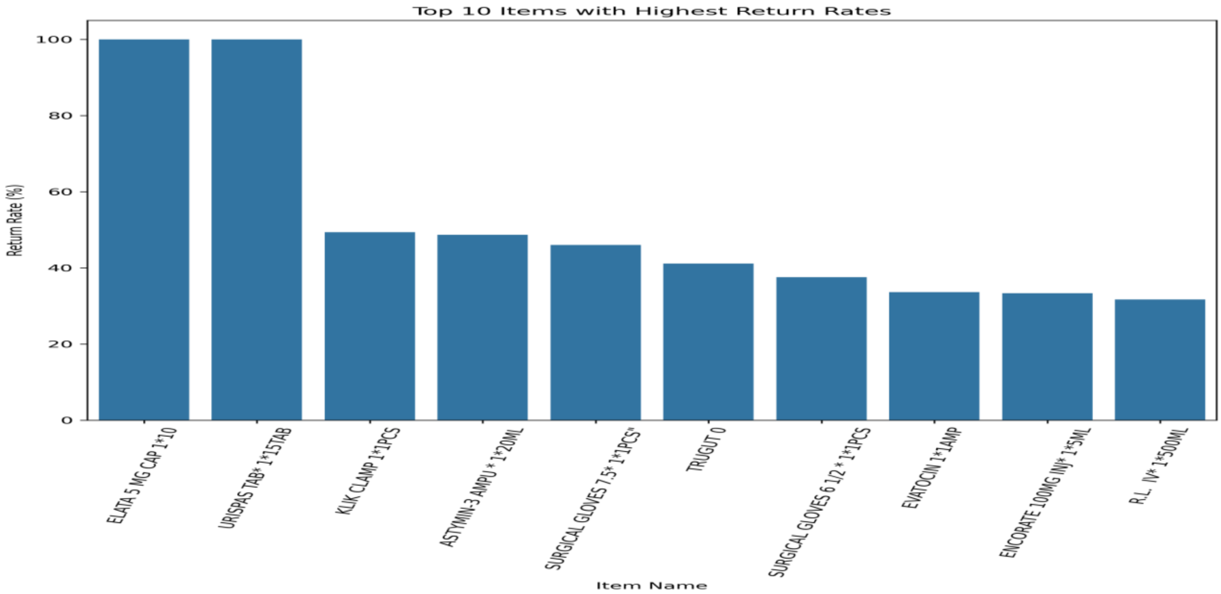


- The ABC (Pareto) analysis shows a highly concentrated revenue structure: about 22% of products generate nearly 80% of total revenue, while the majority of SKUs contribute very little despite occupying shelf space.

- An even smaller core set of items dominates performance, with under 1% of products contributing 15% of revenue and just 0.26% accounting for 5%, confirming a steep, long-tail pattern in the cumulative revenue curve.
- This concentration means that inventory accuracy and availability for a small group of high-impact products are critical, whereas undifferentiated stocking of low-impact items increases holding costs, returns, and obsolescence risk—directly reinforcing the inventory-inefficiency concerns raised in Problem Statement 1.

Finding 6.1.2: Return-Velocity Correlation

Graph 6.1.2: Top 10 Products by Return Rate



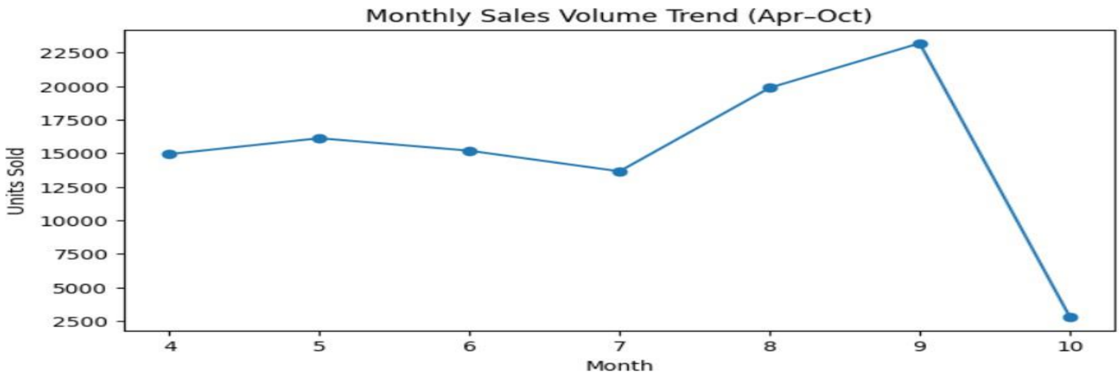
- The top 10 products by return rate show pronounced inventory inefficiency, with two SKUs (ELATA 5 MG CAP and URISPAS TAB) recording 100% returns, meaning the entire procured quantity was sent back without any sales.
- Several others, including KLIK CLAMP, ASTYMIN-3 AMPU, and SURGICAL GLOVES 7.5, have return rates near or above 45%, indicating that close to half of the purchased stock was never absorbed by demand.
- Additional items like TRUGUT 0, SURGICAL GLOVES 6½, and select injectable/IV products also exceed 30% returns, suggesting a systematic pattern of over-ordering rather than isolated mistakes.
- Taken together, these findings indicate an inverse relationship between sales velocity and return rate: slower-moving or specialized items are far more likely to be returned,

confirming inventory inefficiency driven by procurement choices not grounded in actual demand.

6.2.1 Findings for PS2: Seasonality-Driven Demand Misalignment

Finding 6.2.1: Monthly Demand Volatility

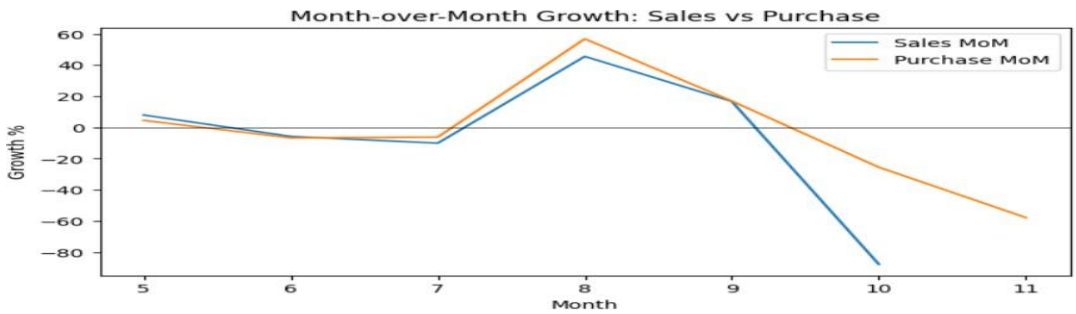
Graph 6.2.1: Monthly Sales Volume Trend



- Monthly sales volumes show clear seasonality: they are relatively stable from April to June, dip in July, and then surge sharply in August and peak in September, indicating a concentrated seasonal upswing rather than gradual growth.
- Such rapid demand acceleration increases the risk of short-term demand–procurement misalignment when buying decisions are based on simple historical averages, directly relating to PS2.
- October appears low only because it is a partial month; its sales are excluded from full seasonal interpretation to avoid misclassifying it as a true post-season decline.

Finding 6.2.2: Procurement-Demand Divergence

Graph 6.2.2: Sales vs Purchase Month-over-Month Growth



- Sales and procurement volumes are strongly aligned overall, but seasonal demand shocks create temporary mismatches due to adjustment lags.
- Month-over-month trends show that procurement generally tracks demand, with sales growth and contraction typically mirrored by corresponding changes in purchase volumes.
- This is supported by a strong positive correlation of 0.86 between sales and purchase MoM growth, indicating largely demand-responsive buying behavior.
- Short periods of sharp seasonal change, however, create timing gaps: purchases do not immediately adjust to abrupt sales drops, causing temporary divergence and highlighting the impact of demand volatility on operational alignment.
- Procurement activity extends beyond the sales horizon, creating end-period misalignment despite strong alignment during peak demand months.

6.3.1 Findings for PS3: Supplier Variability & Procurement Constraints:

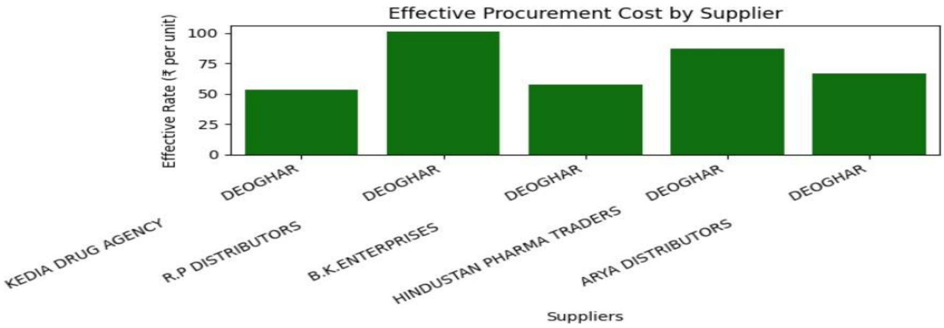
Finding 6.3.1: Supplier Performance Heterogeneity
Graph 6.3.1: Top 5 Supplier Comparison

- Discount percentages are almost uniform across the top five suppliers (=4.75–5%), so headline discounts alone do not meaningfully differentiate supplier value.
- KEDIA DRUG AGENCY is the lowest-cost baseline supplier: it combines large procurement volume with the lowest effective rate, despite offering relatively weaker free-quantity benefits.

- B.K. ENTERPRISES delivers a balanced value proposition, pairing top-tier discount performance with the highest free-quantity score and strong price competitiveness, making it an efficient all-round partner.
- ARYA DISTRIBUTORS and HINDUSTAN PHARMA TRADERS achieve strong price competitiveness despite modest free-quantity support, indicating that their value comes primarily from effective pricing rather than incentive volume.
- Overall, supplier differentiation emerges mainly from free-quantity practices and effective price competitiveness, underscoring the need for a multi-metric scorecard rather than relying on discount percentage alone.

Finding 6.3.2: Cost Optimization Opportunity

Graph 6.3.2: Effective Rate Comparison (Top 5 Suppliers)



- Effective rate analysis shows KEDIA DRUG AGENCY has the lowest median effective procurement rate (₹53.25 per unit), offering about a 47% cost advantage over R.P DISTRIBUTORS (₹100.78 per unit).
- B.K. ENTERPRISES (₹57.71) and ARYA DISTRIBUTORS (₹66.82) occupy a mid-cost band, while HINDUSTAN PHARMA TRADERS (₹87.15) operates at a premium, indicating higher procurement costs.
- Across the top five suppliers, the highest-cost supplier is roughly 89% more expensive than the lowest-cost supplier, demonstrating substantial cost dispersion that invoice rates alone would not reveal.
- Given the current procurement scale (118,311.5 units), even modest per-unit effective cost differences translate into significant total expenditure impacts, reinforcing the need for systematic supplier-level cost benchmarking in procurement decisions.